Nagar Yuwak Shikshan Sanstha's
Yeshwantrao Chavan College of Engineering
(An Autonomous Institution affiliated to Rashtrasant Tukadoji Maharaj Nagpur University)
Hingna Road, Wanadongri, Nagpur - 441 110
NAAC A++
Ph.: 07104-237919, 234623, 329249, 329250 Fax: 07104-232376, Website: www.ycce.edu
Department of Computer Technology

Vision of the Department

To be a well-known centre for pursuing computer education through innovative pedagogy, value-based education and industry collaboration.

Mission of the Department

To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary area by developing Problem-solving skills through emerging technologies.

Session 2025-2026

Vision: *To be a well-known centre for pursuing computer education through innovative pedagogy, value-based education and industry collaboration*	**Mission:** *To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary area by developing Problem-solving skills through emerging technologies.*

Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	**Preparation**	**P: Preparation**	**Pep-CL abbreviation pronounce as Pep-si-lL easy to recall**
PEO2	**Core Competence**	**E: Environment (Learning Environment)**	
PEO3	**Breadth**	**P: Professionalism**	
PEO4	**Professionalism**	**C: Core Competence**	
PEO5	**Learning Environment**	**L: Breadth (Learning in diverse areas)**	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:
Engineering knowledge, Problem analysis, Design/development of solutions, Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World, Ethics, Individual and Collaborative Team work, Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

"I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life." *to contribute to the development of cutting-edge technologies and Research*.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

SHEIKH WASIMUDDIN

Name and Signature of Student and Date
(Signature and Date in Handwritten)

Shiky.
12/03/26

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Session	**2025-26 (EVEN)**	**Course Name**	**JAVA FSD Lab**
Semester	**6**	**Course Code**	**23IOT1607**
Roll No	**63**	**Name of Student**	**SHEIKH WASIMUDDIN**

Practical Number	**05**
Course Outcome	1. Apply OOP concepts to solve the real-life problem. 2. Use the database concepts to integrate in giving OOP oriented solution. 3. Develop full-stack applications integrating frontend, backend, and database with security. 4. Design and develop mini-project with frontend, backend, and database integration, including authentication and testing.
Aim	Swiggy Smart Order Processing System
Problem Definition	As a backend developer at Swiggy in 2026, you are tasked with optimizing the order fulfillment dashboard. The system handles millions of micro-transactions that require real-time validation, status checks, and financial adjustments before they reach the final billing stage. Manual loops and verbose anonymous classes are no longer efficient for this scale of data. Objective: Develop a high-performance order processing module using Java 8 features to streamline bulk order data. The system must process a list of raw order data and generate a finalized billing report while meeting the following specific requirements: 1. Validation (Filtering): From a massive list of transactions, identify and keep only the orders that have been successfully "DELIVERED" Ignore cancelled or pending orders to save processing power. 2. Dynamic Pricing (Mapping): Apply a "Loyalty Discount" of 10% automatically to any order where the bill amount

	exceeds ₹500. This logic must be passed as a functional parameter to keep the code flexible for future seasonal sales. 3. Data Integrity (Optional Handling): Some orders may have missing "Delivery Partner Tips" or optional "Customer Notes" The system must provide default values (e.g., ₹0 for tips) using Optional to prevent NullPointerExceptions and system crashes during peak dinner hours. 4. Reporting (Sorting & Iteration): The finalized list must be sorted by the total bill amount and displayed in a clean, readable format using modern iteration techniques to maximize readability. Technical Constraints: • Avoid traditional for loops and if-else nested blocks to ensure a declarative, functional programming style. • Implement the solution using a combination of Streams, Lambda Expressions, Method References, and the Optional class.
Theory (100 words)	The Swiggy Smart Order Processing System demonstrates the use of Java 8 features to efficiently handle large-scale order data. In high-traffic platforms like Swiggy, traditional loops and conditional statements are inefficient and difficult to maintain. Java 8 introduces functional programming concepts such as Streams, Lambda Expressions, Functional Interfaces, and the Optional class to simplify data processing. Streams are used to filter only delivered orders, apply dynamic loyalty discounts, and sort the final billing data. The Optional class prevents runtime errors by safely handling missing values. This approach improves code readability, performance,and scalability.
Procedure and Execution (100 Words)	Start the program. Create a list of orders containing order ID, order status, bill amount, and optional tip. Define a loyalty discount function that applies a10%discount if the bill amount is greater than ₹500.Use Stream API to filter only the orders with status DELIVERED. Apply the discount function to eligible orders

and add the tip using Optional with a default value of ₹0.
Sort the processed orders in ascending order based on the final bill amount.
Display the sorted order ID and final bill amount using modern iteration(forEach).
Stop the program.

```
import java.util.Arrays;
import java.util.Comparator;
import java.util.List;
import java.util.Optional;
import java.util.function.Function;
import java.util.stream.Collectors;

class Order {

    private int orderId;
    private String status;
    private double billAmount;
    private Optional<Double> tip;
    private Optional<String> notes;

    // Constructor
    public Order(int orderId, String status, double
billAmount, Double tip, String notes) {
        this.orderId = orderId;
        this.status = status;
        this.billAmount = billAmount;
        this.tip = Optional.ofNullable(tip);
        this.notes = Optional.ofNullable(notes);
    }

    // Getters
    public int getOrderId() {
        return orderId;
    }

    public String getStatus() {
        return status;
    }

    public double getBillAmount() {
        return billAmount;
    }
```

```
    public Optional<Double> getTip() {
        return tip;
    }

    public Optional<String> getNotes() {
        return notes;
    }

    // Setter
    public void setBillAmount(double billAmount) {
        this.billAmount = billAmount;
    }

    // Display Order Details
    @Override
    public String toString() {

        double finalTip = tip.orElse(0.0);
        String customerNotes = notes.orElse("No Notes");

        return "Order ID: " + orderId +
                " | Status: " + status +
                " | Bill Amount: Rs." + billAmount +
                " | Tip: Rs." + finalTip +
                " | Notes: " + customerNotes;
    }
}

public class SwiggySmartOrderProcessing {

    public static void main(String[] args) {

        // Raw Order List
        List<Order> orders = Arrays.asList(

                new Order(101, "DELIVERED", 650, 50.0,
"Leave at door"),
                new Order(102, "CANCELLED", 300, null,
null),
                new Order(103, "DELIVERED", 450, null,
"Extra spicy"),
                new Order(104, "DELIVERED", 800, 80.0,
null),
                new Order(105, "PENDING", 550, null, null),
                new Order(106, "DELIVERED", 1200, 100.0,
"Call on arrival")
```

```
        );

        // Loyalty Discount Function (10% discount if bill >
500)
        Function<Order, Order> loyaltyDiscount = order -> {

            if (order.getBillAmount() > 500) {
                double discountedPrice =
order.getBillAmount() * 0.9;
                order.setBillAmount(discountedPrice);
            }

            return order;
        };

        // Stream Processing
        List<Order> finalOrders = orders.stream()

                // Filter delivered orders
                .filter(order ->
order.getStatus().equals("DELIVERED"))

                // Apply discount
                .map(loyaltyDiscount)

                // Sort by bill amount
                .sorted(Comparator.comparing(Order::getBillA
mount))

                // Collect result
                .collect(Collectors.toList());

        // Display Report
        System.out.println("===== Swiggy Final Billing
Report =====");

        finalOrders.forEach(System.out::println);
    }
}
```

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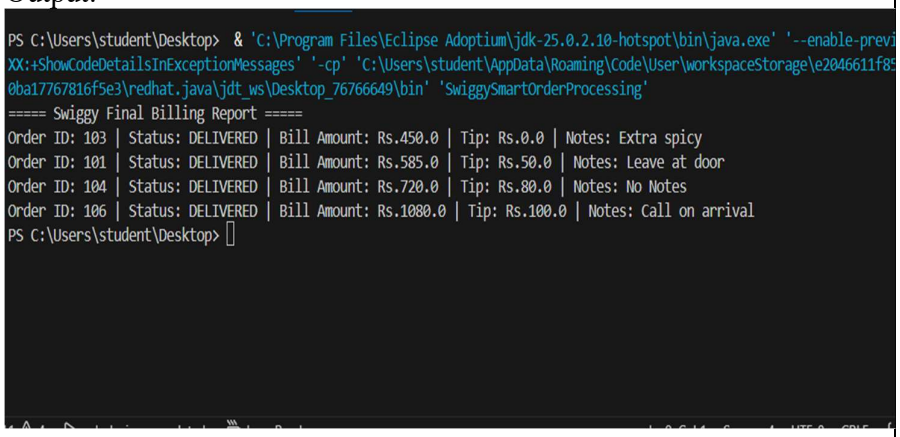
	Output: 
Output Analysis	After executing the program, the system displays a finalized billing report containing only DELIVERED orders. Cancelled and pending orders are automatically excluded during stream filtering
Link of student Github profile where lab assignment has been uploaded	https://github.com/sheikhwasimuddin
Conclusion	The Swiggy Smart Order Processing System successfully demonstrates the effective use of Java 8 features to handle bulk order data in a clean and efficient manner. By using Streams and Lambda expressions, the program avoids traditional loops and complex conditional logic ,resulting in better readability and maintainability. The Optional class ensures data safety by handling missing values without causing runtime errors. Sorting and modern iteration techniques generate a clear and organized billing report. Overall, this practical shows how Java 8 functional programming improves performance, scalability, and code quality in real-world backend systems like **Swiggy**
Plag Report (Similarity index < 12%)	
Date	**12-03-26**

```
PS C:\Users\student\Desktop>  & 'C:\Program Files\Eclipse Adoptium\jdk-25.0.2.10-hotspot\bin\java.exe' '--enable-previ
XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\student\AppData\Roaming\Code\User\workspaceStorage\e2046611f85
0ba17767816f5e3\redhat.java\jdt_ws\Desktop_76766649\bin' 'SwiggySmartOrderProcessing'
===== Swiggy Final Billing Report =====
Order ID: 103 | Status: DELIVERED | Bill Amount: Rs.450.0 | Tip: Rs.0.0 | Notes: Extra spicy
Order ID: 101 | Status: DELIVERED | Bill Amount: Rs.585.0 | Tip: Rs.50.0 | Notes: Leave at door
Order ID: 104 | Status: DELIVERED | Bill Amount: Rs.720.0 | Tip: Rs.80.0 | Notes: No Notes
Order ID: 106 | Status: DELIVERED | Bill Amount: Rs.1080.0 | Tip: Rs.100.0 | Notes: Call on arrival
PS C:\Users\student\Desktop>
```


Paste here Faculty signed observation page.

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Page No.
Date / /

PRACTICAL No:- 05

Aim:- Problem statement for Java 8 features

Problem statement: swiggy smart order processing system.

Output:

- - Swiggy final Billing report - -

Order ID : 103 | status : DELIVERED | Bill Amount : Rs. 450.00 | Tip: Rs. 0.0 | Notes: Extra spicy.

Order ID : 101 | status: DELIVERED | Bill Amount : Rs. 585.00 | Tip: Rs. 50.0 | Notes: No

Order ID: 104 | status : DELIVERED | Bill Amount : Rs. 720.0 | Tip: Rs. 80.0 | Notes: leave at door.

Order ID: 106 | status: DELIVERED | Bill Amount : Rs. 1080.0 | Tip : Rs. 100.0 | Notes: call on arrival.

12/3/26

Shrikrupa